When a Learned σ-Profile Is Not One: Diagnosing the Intermediate in Physics-Embedded Solubility Models

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Abstract

Physics-informed predictors combine learned physical quantities through a fixed thermodynamic relation. The arrangement is attractive because a prediction made through named quantities can be audited, and because a better value for one of them can be looked up and supplied without retraining. We ask whether the values such a network places in the slot of a physical quantity are that quantity. The system is a graph network supplying the σ-profile consumed by COSMO-SAC to predict the solubility of solids, and every measurement is reported against a control. Solubility data do not determine the quantities the relation composes. The network places screening-charge density where quantum chemistry places none, concentrating in the windows that activate the model's hydrogen-bonding term, and the output layer is not the cause. A molecule's profile depends on whether it enters as solute or as solvent, so the model does not return the reference state. The intermediate does not decompose: replacing part of it with reference values is frequently worse than replacing none. Supplying the reference profile improves the activity coefficients the profile exists to produce and degrades the solubility predicted from them.

1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug is formulated as a tablet or an injection, which solvent a crystallisation runs in, and how a product is purified. The value is measured one solute, one solvent and one temperature at a time. Most combinations of industrial interest have therefore never been measured, and a laboratory that needs one is choosing between running an experiment and trusting a prediction. The same is true across process chemistry, where a solvent is selected for a separation long before anyone measures the relevant phase behaviour.

Two kinds of prediction are on offer. A direct model regresses solubility from molecular descriptors or from a representation learned off the molecular graph, and returns a number. A physics-informed model predicts the quantities a thermodynamic argument names, then combines them through a fixed relation taken from that argument. The second kind has become popular for reasons that have little to do with accuracy, on which it usually trails.[1,2] Its internal quantities are things chemistry already knows how to measure and tabulate, so a prediction can be taken apart and checked piece by piece. Each piece can be trained against reference data that a direct regressor has no way to use. And the fixed relation supplies, for free,

structure that a purely statistical model would have to learn from data it does not have, which is the situation whenever the molecule in question resembles nothing in the training set.

That design rests on an assumption so natural it is rarely written down. If the quantity inside the model is a real physical quantity, then a better value for it can be looked up and handed to the model, and the prediction should improve. The assumption is what makes the internal quantity worth having, because it is the difference between a model that can be audited and repaired piece by piece and an elaborate curve fit with suggestive variable names. It also carries direct practical weight: it is the reason a laboratory would pay for a quantum-chemical calculation on a new compound, or curate a table of measured melting properties, rather than spend the same effort on more solubility measurements.

The quantity this work examines is the σ-profile. In the conductor-like screening model for real solvents and in its segment-activity reformulation COSMO-SAC,[3] a molecule enters the thermodynamics only through the distribution of screening-charge density over its cavity surface, that is, through how much of that surface carries charge density of each value. Everything else about the compound is discarded. The distribution comes from a quantum-chemical calculation with a continuum solvation treatment and is tabulated once per molecule.[4] It is an unusually attractive thing to learn: it is a fixed-length vector, it is tabulated for thousands of compounds, and supplying it from structure removes the only expensive step in the pipeline.

Group-contribution schemes supplied it first,[5,6] and learned predictors followed, taking the profile from a line notation or a molecular graph.[7–9] More recent work embeds the learning deeper, replacing both the profile and the segment-activity calculation with networks trained inside the thermodynamic structure, and reports accuracy beyond the model it is built from.[10] Others keep the thermodynamics and dispense with the named intermediate altogether,[11] or learn a thermodynamic potential directly from phase behaviour. Across this literature the internal quantity is described as physically meaningful on the grounds that it appears in the thermodynamic formulation. Validation, where it is reported for the intermediate at all, compares averaged predicted profiles against averaged reference profiles, or rests on the accuracy of the final prediction.

What has not been asked is whether the values a network places in that slot are the physical quantity. The question is not rhetorical. A solubility measurement constrains the combination of the intermediate with the fixed relation, not the intermediate on its own, so different internal values can reproduce the same measurements equally well; the directions the data leave undetermined are where a flexible network is free to move. Machine learning has met this before under a different name. In models that predict human-meaningful concepts and then predict the target from the concepts alone, the concepts leak. They carry information their labels do not name, and interventions on them behave in ways the names do not predict.[12–15] The thermodynamic case differs in one respect that makes it tractable. A σ-profile is not a human label but a computable quantity, so the leak can be measured against the quantity itself.

This work measures it. We examine a pipeline that predicts the solubility of a solid through COSMO-SAC, with a graph network supplying the σ-profile, trained both against tabulated reference profiles and through the closure. Solubility data do not determine the quantities the model composes. The network places screening-charge density where the reference carries none, and does so preferentially in the windows the hydrogen-bonding term reads. The output layer is not the cause: the same network reaches far smaller values elsewhere in the same profile, and an untrained network of the same architecture is worse by more than an order of magnitude. A molecule's profile depends on whether it enters as solute or as solvent, so the predicted activity of a molecule in itself departs from unity. What misplaced density costs the prediction varies by more than an order of magnitude along the charge-density axis and steps at the hydrogen-bonding threshold. The

learned profile does not decompose: replacing part of it with the reference values is frequently worse than replacing none. And the error that reaches the thermodynamics lives in the magnitude of the profile where it is populated, not in the spurious density where it should be empty.

This article is organised as follows. Section 2 describes the thermodynamic relation being closed, the network that supplies its input, the reference profiles and the data. Section 3 reports the measurements, each against the control that would refute it. Section 4 draws out what they imply for the design of models of this kind, and Section 5 states the limits of the evidence. Section 6 summarises.

2 The system under study

2.1 The relation being closed

The solubility of a crystalline solid in a liquid follows from a balance between two quantities. Breaking the crystal costs an amount set by the melting temperature T_m and the enthalpy of fusion ΔH_{fus}; mixing the freed molecules into the solvent is governed by the activity coefficient γ_2 of the solute in that solvent. For an ideal solution the two combine into the mole fraction at saturation,

$$\ln x_2 \;=\; -\,\Phi(T) \,-\, \ln \gamma_2, \qquad \Phi(T) \;=\; \frac{\Delta H_{\mathrm{fus}}}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right), \tag{1}$$

with R the gas constant. The first term is the crystal contribution and carries no information about the solvent; the second is where the chemistry of the mixture enters. Because γ_2 itself depends on x_2, equation (1) is solved rather than evaluated.

COSMO-SAC[3] supplies γ_2. It represents each molecule by its σ-profile: the area of the molecular cavity surface carrying screening-charge density σ, discretised onto a fixed grid of 51 bins spanning roughly $\pm 0.025\, e^{-2}$. A molecule's identity enters the model through this vector and through the total cavity area; nothing else about the compound is used. Segments exchange with a common pool, and a self-consistent calculation returns a segment activity coefficient for each bin, from which the molecular activity coefficient follows. The exchange energy contains a term that switches on only for segments beyond a threshold σ_{hb} in magnitude, which is how the model represents hydrogen bonding; a segment qualifies by its charge density alone, with no reference to which atom carries it. That threshold divides the charge-density axis into a core and two outer windows, and the division recurs throughout what follows.

2.2 The network that supplies the profile

A graph neural network reads the solute and the solvent and produces, for each, a σ-profile and a cavity area; these are passed to the COSMO-SAC calculation and to the solver for equation (1). A second head predicts the crystal properties T_m and ΔH_{fus} that enter $\Phi(T)$. The profile head is a feed-forward network on the per-molecule representation, ending in a normalised, non-negative vector of 51 bins. The same head serves both roles, but reads a role-dependent representation, so a compound receives one profile as solute and another as solvent.

The head is trained in two ways at once, and this is the property that makes the pipeline suitable for the present question. It is supervised directly against tabulated reference profiles, as a predictor of those profiles would be. It is also trained through the thermodynamic relation, so the gradient of the solubility loss reaches it. The nearest prior model instead supervises its profile predictor externally and then freezes it, so that no gradient from the phase-equilibrium data reaches the intermediate.[10] The pipeline studied here therefore occupies the position where both influences act, which is where their relative effect can be read.

2.3 Reference profiles, data and arms

Reference σ-profiles are taken from the VT-2005 tabulation,[4] computed by quantum chem-

istry with a continuum solvation treatment. Throughout, "reference" means a value from that table, and departures are measured against it. How far the tabulation itself depends on the underlying electronic structure method is a separate question[16] and is not at issue here, since every comparison in this work uses one table.

Solubility data are drawn from a curated compilation of measured solid solubilities,[17] split so that no scaffold[18] appears in both training and test sets, which is the harder of the two standard settings and the one relevant to new compounds. Infinite-dilution activity coefficients used to score the closure directly come from an independent compilation. All results below are reported over five training seeds; the arms differ only in the seed, so the spread across them is the natural scale against which any effect must be read. Architectural details, hyperparameters, the construction of the splits and the auxiliary supervision streams are given in the Supporting Information.

2.4 How the measurements are read

Each measurement in Section 3 is a contrast, and each is reported against a control chosen so that it would fire if the instrument responded to something other than the intended effect. Where a quantity is compared with a reference table, the control is the same quantity computed on profiles that carry the same total area but no chemistry. Where part of a profile is replaced, the control is a part of the same size chosen at random. Where an effect is attributed to training, the control is an untrained network of the same architecture. The reading rule for each measurement, together with the result that would refute it, was fixed before the measurement was made; the rules and their outcomes are listed in the Supporting Information. Table 1 collects every test with the control beside it, because a result of this kind is not interpretable without one.

3 Results

Table 1 lists every measurement reported below together with the control it is read against. The order follows the question rather than the chronology: first whether the data determine the internal quantity at all, then what the network does with the freedom that remains, then what that costs the thermodynamics.

3.1 Solubility does not determine the quantities the model composes

Equation (1) is a sum of two terms, and a solubility measurement constrains the sum. The question is whether a set of such measurements separates them. For a fit of the four parameters that govern the crystal term and the temperature dependence of the activity coefficient, the information a solubility series carries is strongly concentrated: the smallest eigenvalue of the information matrix is five orders of magnitude below the largest. The consequence is quantitative rather than abstract. No estimator can determine the enthalpy of fusion from such a series to better than about $\pm 20\ \mathrm{kJ\,mol^{-1}}$, which is wider than the range the quantity takes across the compounds in the data. Supplying the melting temperature and the enthalpy of fusion as independent labels reduces that bound by an order of magnitude.

The crystal enthalpy and the temperature slope of the activity coefficient are collinear because each enters equation (1) through $1/T$. A fit can trade one against the other and reproduce the measurements equally well. The same argument applies with more force to the σ-profile, which enters through fifty-one numbers rather than one. A solubility measurement constrains what the profile produces after the closure has acted on it. It leaves the profile itself undetermined in every direction the closure does not expose. What the network places in those directions is not decided by the data, and the remainder of this section measures what it places there.

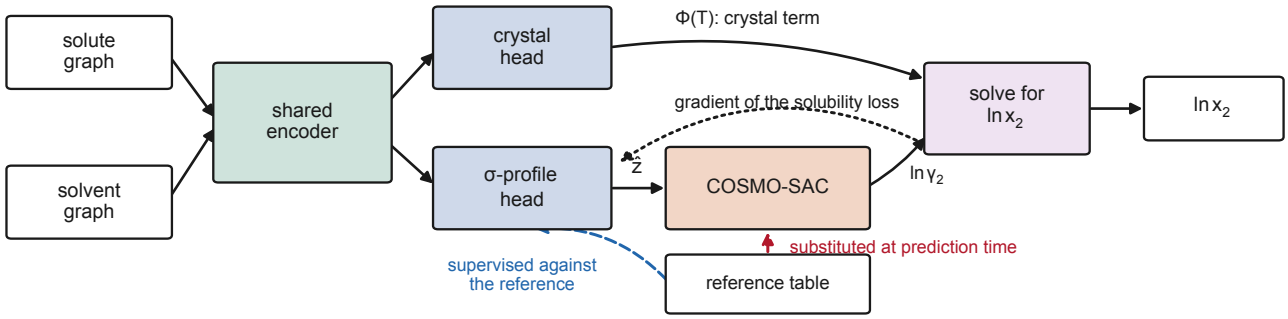


Figure 1: The pipeline. A shared encoder reads the solute and solvent graphs; one head supplies the crystal term of equation (1) and another the σ-profile $\hat{z}$ consumed by COSMO-SAC. The profile head is supervised against the reference table and also receives the gradient of the solubility loss. At prediction time the reference profile can replace the learned one.

3.2 The network places charge density where the reference carries none

The reference tabulation is sparse by construction. A molecule with no hydrogen-bond donor has no cavity surface carrying donor-side charge density, and the tabulated profile is exactly zero there. Across the compounds examined here, well over half of the reference profile's bins are exactly zero.

The learned profiles are not sparse anywhere. Figure 2 shows the donor-window area for sixteen common solvents. Twelve of the sixteen are exactly empty in that window in the reference; all sixteen carry area there in the learned profile. The amounts are small as a fraction of the molecule's surface, between a tenth of a percent and about five percent, and the hydrogen-bonding term reads them regardless, because a segment qualifies for that term by its charge density alone.

Where the misfit lands is not uniform across the charge-density axis. Take the residual between the learned and the reference profile, and express it relative to the share of surface area each window holds. The donor window is then over-represented by about a factor of four and the acceptor window by a factor of two, while the nonpolar core is under-represented. The displacement is therefore oriented: the network's departure from the tabulation concentrates in exactly the regions that switch on the model's hydrogen-bonding term, and thins in the region that does not.

3.3 The output layer is not the cause

An obvious explanation is architectural. The head ends in a normalisation that cannot return an exact zero, so a profile that should be empty in a bin will instead be slightly populated there, and the appearance of a defect would be an artefact of the parameterisation. The explanation is testable, and it fails.

Figure 3a shows, for every bin in which the reference is exactly zero, how much mass the trained head actually places there. The distribution reaches 10^{-11}, and between a third and a half of those bins already lie below 10^{-5}. A layer that can reach 10^{-11} is not what holds the remaining bins at 10^{-5}. Figure 3b makes the same point from the other direction: an untrained network of the same architecture places about 0.57 of the profile's mass in the bins where the reference is empty, and training reduces that to about 0.02. The residual population is something training arrives at, not something the parameterisation imposes.

This matters beyond the present pipeline. A recent physics-embedded model uses a rectified output with the same normalisation and can therefore emit exact zeros.[10] The measurement

Table 1: The diagnostic battery. Each test is read against the null named beside it; a result without its null is not a result. Numbers are medians over five training seeds unless a range is given.

Question	Instrument, and the null it is read against	Result	Reading
Does solubility determine the factors it composes?	Fisher information of the four-parameter fit; Cramér–Rao bound on the enthalpy of fusion, against the same bound with external labels supplied	condition number 2.3×10^5; bound $\pm 20\,\mathrm{kJ\,mol^{-1}}$, falling to ± 1.7 with the labels	No. The crystal enthalpy is collinear with the activity's temperature slope.
Does the head use the freedom?	residual of learned minus reference profile by charge-density window, 71 molecules, against each window's share of surface area	donor window 4.0×, acceptor 2.0×, nonpolar core 0.88×	Yes, and into the windows the hydrogen-bonding term reads.
Is the output layer responsible?	mass in bins where the reference is exactly zero, against the smallest bin the same head reaches anywhere and against an untrained head	head reaches 10^{-11}; 34–51% of those bins already below 10^{-5}; pedestal 0.02–0.03 against 0.57 untrained	No. The pedestal is learned. 58% of reference cells are exactly zero.
Is the reference state preserved?	activity of a molecule in itself, against the one-profile control that must return zero identically	control 0 exactly; median $\lvert \ln \gamma \rvert$ 0.012–0.044, largest 0.80	No. A molecule's two role profiles do not coincide.
What does misplaced mass cost, and where?	mean $\lvert \Delta \ln \gamma \rvert$ from moving a fixed mass into each bin, against random profiles of the same area	edge-to-core ratio 21× and 15× on the two row sets, against 1.6× for random profiles	Strongly non-uniform, with a step at the bonding threshold.
Does the intermediate decompose?	reference shape substituted into part of the grid, against a randomly chosen part of the same size	the whole shape helps on 5 of 5 seeds; a random third is worse than the fully learned profile in 94% of draws	No. The parts do not add, so per-bin attribution is undefined.
Where does the error that reaches the closure live?	reference shape in the bins where it is nonzero, against exact zeros in the bins where it is zero	+89% of the gap recovered against −9%	In the magnitude of the occupied bins, not in the pedestal.

above indicates that the difference between the two output layers is not what decides whether the learned profile resembles the tabulated one.

3.4 A molecule is not ideal in itself

Any activity coefficient model must return unity for a pure component: a molecule surrounded by copies of itself is, by definition, in its reference state. In COSMO-SAC the property holds identically, because the residual term is a difference between a segment's activity in the mixture and in the pure compound, and the two coincide when one profile is used on both sides. Supplying a single profile to both sides of the calculation returns exactly zero here, which confirms that the implementation carries the property.

The pipeline does not. Because the head

reads a role-dependent representation, a compound receives one profile when it enters as solute and a different one when it enters as solvent. Pairing a molecule with itself therefore pairs two different profiles, and the activity coefficient departs from unity. The median departure is a few hundredths in $\ln \gamma$ and the largest exceeds 0.8. The violation is a property of the learned intermediate rather than of the thermodynamic relation, and it is invisible in any evaluation that never pairs a compound with itself, which is every evaluation on solubility or on mixture data.

3.5 What misplaced density costs, and where

The preceding measurements establish that density sits where the reference says it should not. They do not say what that costs, and the cost is not uniform. Figure 4a reports, for each bin, the mean change in the predicted activity coefficient produced by moving a fixed amount of mass from the core into that bin. Moving mass within the core is almost free. Moving the same mass into the outer windows is more than an order of magnitude more expensive. The ratio between the two is far larger than it is for random profiles of the same total area, so the structure reflects the chemistry of the exchange rather than the geometry of the grid.

The cost does not grow smoothly with distance along the axis. It is flat across the core, rises steeply as the hydrogen-bonding threshold is crossed, and then saturates: once mass is inside the bonding region, pushing it further out changes the prediction comparatively little. The consequence is shown in Figure 4b. Reading the measured cost as a potential and taking its gradient gives the weight under which the usual transport distance between two profiles would price their difference correctly. That weight peaks at the boundary, not deep in the outer window. A treatment that weighted the far corner most heavily, which is what the geometry of the grid would suggest, would weight the wrong place.

3.6 The intermediate does not decompose

If the learned profile were a σ-profile carrying a per-bin error, then correcting part of it should help in proportion to the part corrected. It does not. Replacing the learned shape with the reference shape over the whole grid improves the predicted activity coefficients on every seed. Replacing it over a randomly chosen third of the bins makes the prediction worse than leaving the profile entirely alone in a fifth of the draws. The outcome ranges from a large improvement to a loss several times the size of the whole gap being closed.

A profile assembled from correct values in some bins and learned values in the rest is not intermediate between the two. The segment-activity calculation is a self-consistent solution over the whole distribution, so an internally inconsistent profile can sit far from both of the profiles it was built from. Two things follow. The first is methodological: the share of the error attributable to any subset of bins is not defined on this surface, so an analysis that partitions the grid and attributes error to the parts cannot be carried out, whatever the partition. The second concerns the premise under test. Partial correctness of the intermediate is not partially useful; it is frequently worse than consistent incorrectness.

3.7 The error that reaches the thermodynamics is in the occupied bins

One partition is well posed, because both of its parts are substituted in full: the bins where the reference is nonzero, and the bins where it is zero. Figure 5a reports both. Supplying the reference values in the bins the reference actually occupies recovers close to ninety percent of the gap between the learned profile and the reference, on every seed. Setting exact zeros in the bins the reference leaves empty recovers nothing: the mean is slightly negative and the sign changes from seed to seed, and the result sits inside the spread of a control that zeroes the same number of randomly chosen cells.

The spurious density is therefore not what the thermodynamics is responding to. The error that propagates through the closure lives in the magnitude of the profile where the profile is populated. The cost map of Section 3.5 does not localise this error either. Figure 5b shows that substituting the reference into the third of the grid the closure is most sensitive to recovers no more of the gap than substituting into a randomly chosen third, and that the sign varies across seeds. Sensitivity per unit of displaced mass and total contribution to the error are different quantities, and the core carries most of the mass.

Separating the profile's shape from the molecule's cavity area, which COSMO-SAC consumes as a separate scalar, splits the gap unevenly: about two thirds of it is carried by the shape and the remainder by the area.

4 Discussion

The measurements describe a quantity that occupies the place of a σ-profile without being one. It is sparse where the reference is dense and populated where the reference is empty. It depends on the role the molecule plays in the pair, so the model it feeds does not return the reference state. And it does not decompose, so the familiar move of correcting part of it and expecting part of the benefit is unavailable.

The sharpest statement of the situation is a contrast between two substitutions of the same values. Supplying the reference profile improves the quantity the profile exists to produce: the predicted activity coefficients move most of the way toward the measured ones, on every seed. The same substitution degrades the solubility the pipeline is built to predict. Both facts are measured on the same checkpoints and the same compounds. Taken together they say that the downstream parts of the model have adapted to the profile the network supplies rather than to the one the physical argument names. Handing the model a physically better value therefore moves it away from the operating point it was fitted at. This is the behaviour the concept-bottleneck literature reports for learned concepts in general,[12,13] arrived at here through a thermodynamic decoder and, because the concept has a computable ground truth, measured against the concept itself rather than inferred from how interventions behave.

It follows that the assumption set out in the Introduction is not a property of physics-informed models as such but a property of how the internal quantity is trained. The design choice that matters is whether the gradient of the target reaches the intermediate. Where it does not, as in a model whose profile predictor is supervised externally and then frozen,[10] the intermediate remains what it is named after, to within its own fit error, and substitution is a sensible operation. Where it does, the intermediate becomes a free internal coordinate that happens to be initialised and regularised toward a physical quantity, and the operations the physical reading licenses stop being safe. The present pipeline sits between the two, with both influences acting, and the measurements show which one wins.

For practice this suggests a different division of labour than the one the literature has settled into. If the internal quantity is to be read as chemistry, exchanged with tabulated values, or audited against reference data, it should be supervised against that reference and then held fixed. The capacity needed to fit the data belongs somewhere that makes no physical claim. If instead the intermediate is allowed to move under the target's gradient, the gain in accuracy is real but the physical reading has to be dropped, and with it the argument that recommends these models over direct regression in the first place. What is not available is to train the intermediate end to end and retain the interpretation, which is the position most of this literature currently occupies.

Three measurements in the battery are worth separating from the rest because each removes an explanation that would otherwise be reached for. The output layer is not responsible for the spurious density: the same head reaches values six orders of magnitude smaller elsewhere in the same profile. The cost map does not localise the error: the bins the closure is most sensitive to



Figure 2: Donor-window area for sixteen solvents in the reference tabulation (left) and in the learned profile (right). Percentages give that window's share of the molecule's cavity surface in the learned profile.

account for no more of the gap than a randomly chosen third. Sensitivity is measured per unit of displaced mass, and most of the mass sits in the insensitive core. And the spurious density in the empty bins, the most visible defect and the one easiest to describe, is not what the thermodynamics responds to; the error that propagates is in the magnitude of the profile where the profile is populated.

The finding is conditional on the fixed relation being an adequate description of the chemistry it is applied to. Where a relation is well specified for its domain, supplying it with more accurate inputs behaves as the physical reading predicts. A second system, in which a linear free-energy relation stands in for the closure and its adequacy varies by chemical stratum, separates in that way; it is reported in the Supporting Information. The result here should therefore be read as a statement about an intermediate trained through an approximate closure, which is the common case, rather than as an argument against building physics into a model.

5 Limitations

The evidence comes from one pipeline, one closure and one reference tabulation. Every departure reported above is measured against the VT-2005 table. How far that table itself moves with the electronic structure and solvation treatment behind it is a separate question.[16] A different tabulation would shift the reference against which the learned profile is scored, but not the comparisons between arms, which all use one table.

This work does not report a more accurate solubility model. On the scaffold-disjoint split used throughout, the physics path trails a matched direct control sharing the same encoder in the mean over seeds, and both sit near a random-forest baseline. Both are well be-

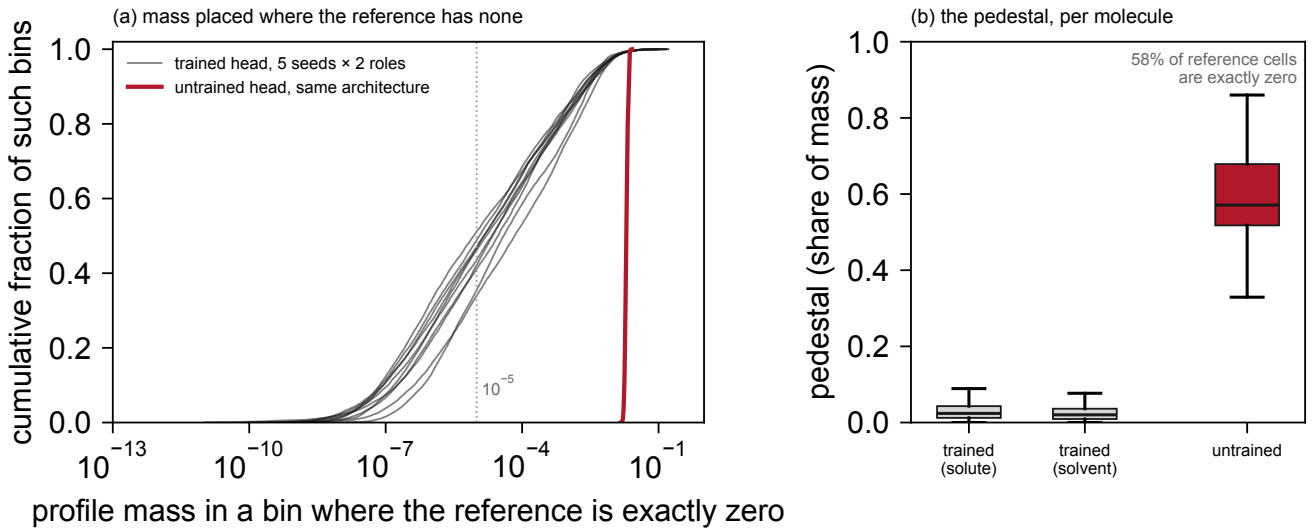


Figure 3: (a) Cumulative distribution of the profile mass placed in bins where the reference is exactly zero, for five seeds in both roles and for an untrained network of the same architecture. (b) The same mass summed over each molecule.

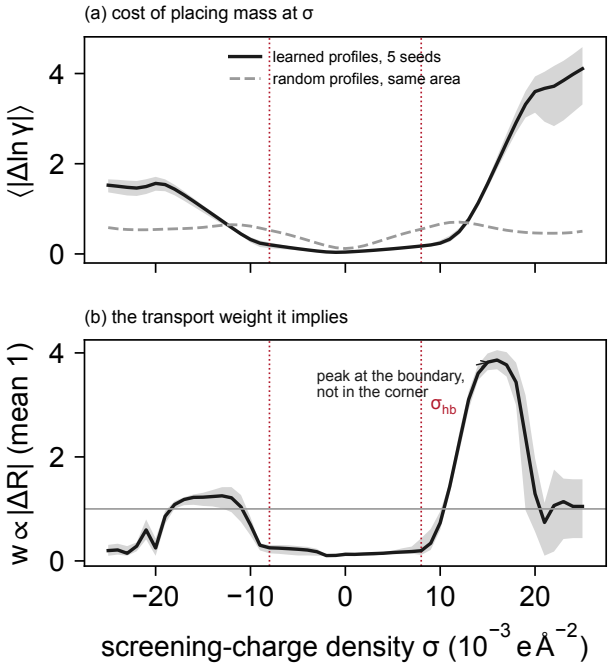


Figure 4: (a) Mean change in the predicted activity coefficient produced by moving a fixed mass from the core into each bin, for five seeds (band) and for random profiles of the same total area. (b) The gradient of that cost, normalised to mean one. Dotted lines mark $\pm\sigma_{\mathrm{hb}}$.

hind the leading direct models on extrapolation to new solutes,[1] and absolute errors are within reach of the disagreement between independent measurements of the same system. No published model was retrained here, so the comparison with prior work is a comparison of design choices rather than of performance.

Results are reported over five training seeds, and for some quantities that is not enough. The cost of the spurious density, expressed in the error of the predicted activity coefficients, varies across seeds by several times the spread of the arms themselves. Its sign is therefore not determined by the present evidence. What is determined is that it is not large and positive. The asymmetry of the cost map between the two sides of the charge-density axis depends on which set of molecule pairs it is computed over, and is reported for both.

The closure is the 2002 formulation of COSMO-SAC with a single hydrogen-bonding term, run without the combinatorial contribution, which requires molar volumes the pipeline does not supply. Later formulations split the hydrogen-bonding term by segment type and would change where on the charge-density axis the cost steps, though not the existence of a step. The segment calculation is iterated to a fixed count; the count used at training and at scoring is recorded with each run, because the two need not agree and the predicted activity coefficients depend on them.

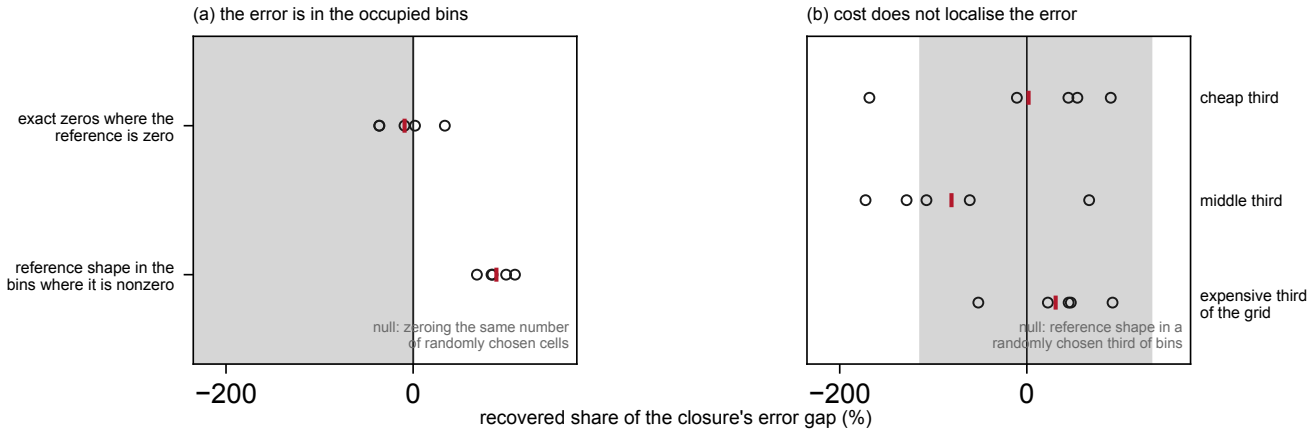


Figure 5: Share of the gap between the learned and the reference profile recovered in the predicted activity coefficients, one circle per seed and a bar at the mean. (a) Reference values supplied in the bins where the reference is nonzero, and exact zeros in the bins where it is zero. (b) Reference values supplied in thirds of the grid ordered by the cost of Figure 4. The shaded band is the central 90% of the control named in each panel.

Finally, the measurements concern substitution at prediction time on an intermediate that was trained through the closure. They do not establish what would happen in a model whose intermediate is supervised externally and frozen. That design is the one the results recommend, and testing it requires training such a model rather than measuring this one.

6 Conclusions

A network trained inside a fixed thermodynamic relation fills the relation's input slot with values that are not the physical quantity the slot names. The learned σ-profile places screening-charge density where quantum chemistry puts none, concentrating in the windows that trigger the model's hydrogen-bonding term. It differs depending on whether the molecule is treated as solute or solvent, so a molecule is not predicted to be ideal in itself. And it cannot be corrected in parts: replacing a fraction of it with reference values is often worse than replacing none. The output layer is not the cause, and the error that reaches the thermodynamics lies in the magnitude of the profile where it is populated rather than in the spurious density where it should be empty.

Supplying the reference profile improves the activity coefficients the profile exists to produce and degrades the solubility the pipeline predicts. The interpretation that makes physics-informed models attractive, that a better value for an internal quantity can be looked up and handed to the model, holds only where that quantity is supervised externally and shielded from the target's gradient. Where the intermediate is trained end to end, it is an internal coordinate with a physical name, and the operations the name licenses should be measured before they are used.

Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review & editing.

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Notes

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Data and Software Availability

The analysis code and the intermediate artifacts small enough to version are available at `https://github.com/doctawho42/tgnn-solv` under the MIT licence; the commit the reported numbers were produced from is recorded in the deposit. Model checkpoints, the processed split and the per-arm prediction files are deposited in the Zenodo archive `10.5281/zenodo.22263434`. The deposit covers the COSMO-SAC arms reported here; weights for the NRTL variants discussed in the Supporting Information were not retained. Solubility data are BigSolDB 2.0;[17] reference σ-profiles are the VT-2005 database[4] as redistributed with Ref.;[19] infinite-dilution activity coefficients are from ThermoML;[20] the acid-dissociation data used in the Supporting Information are the OPERA compilation.[21]

Supporting Information Available

Architecture and training details; construction of the splits and the auxiliary supervision streams; the pre-declared reading rule for each measurement with its outcome; the decomposition of the model's error against what its inputs leave unresolved; and the second system based on a linear free-energy relation.

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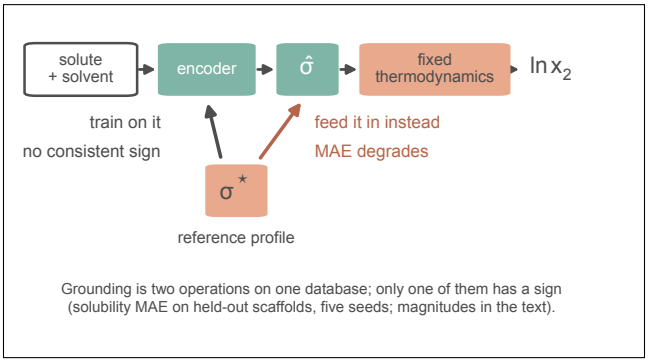
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TOC Graphic


solute
+ solvent
encoder
$\hat{\sigma}$
fixed
thermodynamics
$\ln x_2$
train on it
no consistent sign
feed it in instead
MAE degrades
$\sigma^\star$
reference profile
Grounding is two operations on one database; only one of them has a sign
(solubility MAE on held-out scaffolds, five seeds; magnitudes in the text).